

ShowTime OTT Services

Business Analysis Report

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Project Background and Objective

OTT platforms like Netflix, Hotstar, Amazon Prime, etc., have changed the way people watch movies and shows. Now most people like to watch content online at any time instead of watching TV. Because of this, OTT platforms are growing very fast, and competition is also increasing between them.

In this project, I worked as a Data Scientist for a platform called **ShowTime**. They wanted to know what affects the number of people who watch a show or movie on the **first day** of its release. This is very important for them because if a content performs well on the first day, then it can become popular later also.

Some of the things that can affect the viewership are:

- Number of people visiting the app or website
- How many ads were shown for the content
- What day the content was released
- What season it was
- If any major sports match was going on that day
- What is the genre
- How many people saw the trailer

The company gave us a dataset with these details, and my job was to study the data, do some analysis, and then build a **Linear Regression model**. The model will help to understand which things actually impact first-day views the most. After that, I also gave some suggestions to the company based on the results so that they can plan better in the future.

Understanding the Data

The dataset I got from ShowTime had information about 1,000 different content releases on their platform. Each row in the dataset represents one piece of content, like a show or movie.

There were 8 columns in the data:

- visitors: This shows the average number of users (in millions) who visited the platform in the last 7 days before the release.
- ad_impressions: This tells how many ads (in millions) were shown for this particular content.
- major_sports_event: This tells if there was any big sports event happening on the same day. It has values 1 for yes and 0 for no.
- genre: This is the type or category of the content, like Comedy, Horror, Thriller, etc.
- dayofweek: The day on which the content was released, like Monday, Friday, etc.
- season: The season in which the content was released, such as Summer, Winter, Fall, or Spring.
- views_trailer: This tells how many people (in millions) watched the trailer of the content.
- views_content: This is the target column. It tells how many people (in millions) watched the content on the first day of release.

The shape of the dataset was 1000 rows and 8 columns. There were no missing values in the data. Some columns like genre, season, and dayofweek were in text format, so we later changed them into a format that the model could understand. Overall, the data was clean and ready for analysis.

Exploratory Data Analysis (EDA)

In this part of the project, I tried to understand the dataset better by looking at each column, finding patterns, and checking how different features are related to our target column `views_content`, which is the number of first-day views.

This also helped me answer the questions given in the project and find out which features might be important when building the model.

3.1 Shape, Data Types, and Missing Values

- The dataset has 1000 rows and 8 columns.
- There were no missing values in any column.

The column types were as expected:

Numerical: `visitors`, `ad_impressions`, `major_sports_event`, `views_trailer`, `views_content`

Categorical: `genre`, `dayofweek`, `season`

This means the data was clean and didn't need any missing value treatment.

3.2 Summary Statistics

I used `.describe()` to look at the summary of numeric columns.

Some points I observed:

- `visitors`: Mean = 1.70 million, Min = 1.25, Max = 2.34
- `ad_impressions`: Mean = 1434 million, Range = 1010 to 2424
- `views_trailer`: Mean = 66.91 million, Max = around 200 million

- views_content: Mean = 0.47 million, Range = 0.22 to 0.89 million

These helped me understand the general spread of values and also gave me clues about possible outliers.

In this section, we were given a dataset related to football players, and we had to answer a few probability questions using simple data counts and formulas. Let's look at each question one by one.

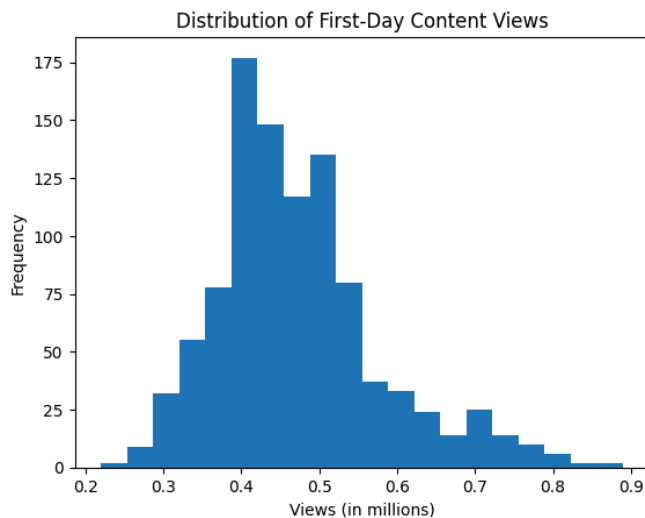
	visitors	ad_impressions	major_sports_event	views_trailer	views_content
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	1.704290	1434.712290	0.400000	66.91559	0.473400
std	0.231973	289.534834	0.490143	35.00108	0.105914
min	1.250000	1010.870000	0.000000	30.08000	0.220000
25%	1.550000	1210.330000	0.000000	50.94750	0.400000
50%	1.700000	1383.580000	0.000000	53.96000	0.450000
75%	1.830000	1623.670000	1.000000	57.75500	0.520000
max	2.340000	2424.200000	1.000000	199.92000	0.890000

3.3 Univariate Analysis

I checked the individual columns using bar plots and histograms.

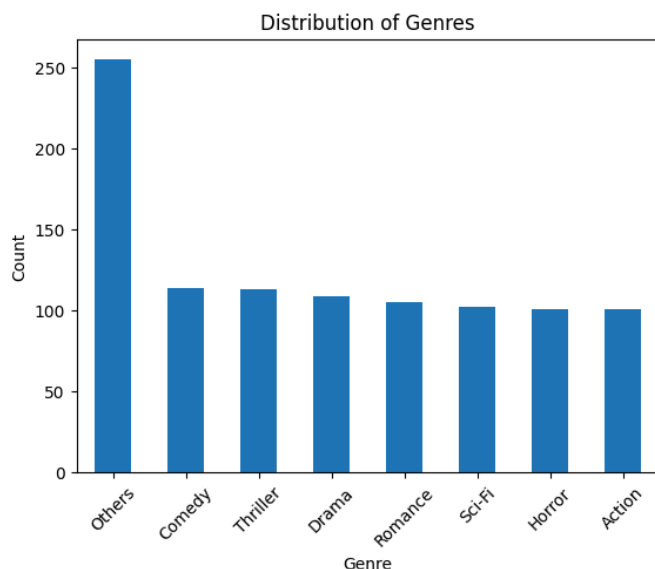
a) Distribution of First-Day Content Views

Most of the values for views_content were between 0.4 and 0.5 million.



b) Genre Distribution

Almost all genres had similar numbers, around 100 entries. The "Others" category had the highest — around 250 entries. This could mean a lot of mixed or niche content was grouped there.



c) Day of the Week

Content was released on all days, and the distribution was even. This shows that the company experiments with different release days.

d) Season of Release

All four seasons were present. Summer had slightly more average views, while Fall had slightly fewer.

3.4 Key Questions

1. What does the distribution of content views look like?

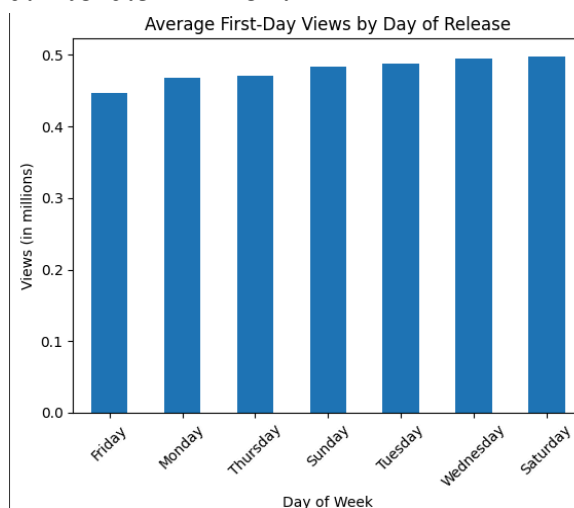
Most shows and movies get between 0.4 to 0.5 million views on the first day. Very few go above or below that.

2. What does the distribution of genres look like?

Genres like Comedy, Drama, Horror, Thriller, etc., all had close to 100 shows each. But the "Others" category had the highest count, with around 250. This might mean that many unique or uncategorized content types are released often.

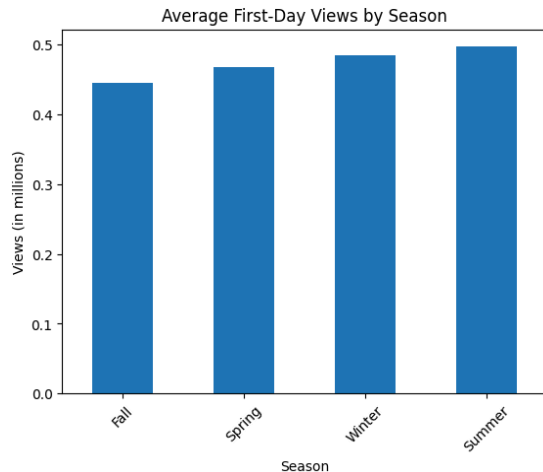
3. How does the viewership vary with the day of release?

From the graph, it was clear that Saturday and Wednesday releases did slightly better on average. Other days had similar view counts between 0.4 to 0.5 million.



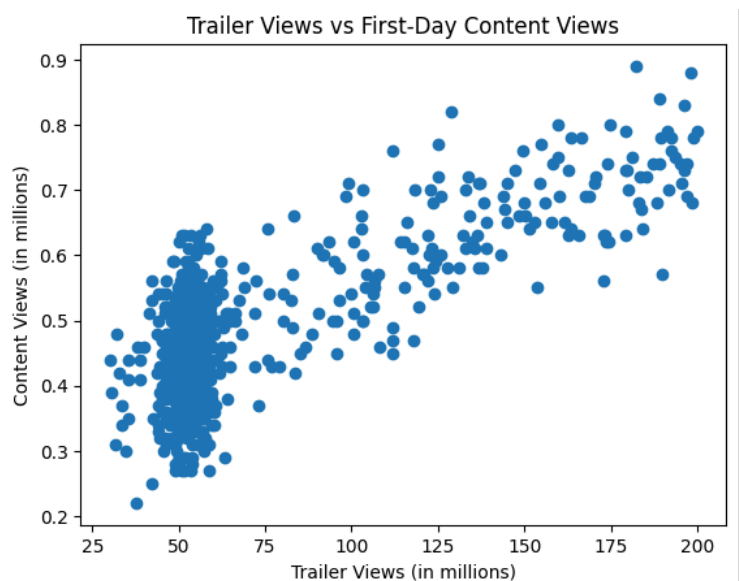
4. How does the viewership vary with the season of release?

Summer releases had the highest average views, followed by Spring. Fall had the lowest. This might be due to school vacations, holidays, or user behavior.



5. What is the correlation between trailer views and content views?

The correlation between `views_trailer` and `views_content` was found to be 0.75, which is a strong positive correlation. It makes sense — the more people who watch the trailer, the more likely they are to watch the actual content.

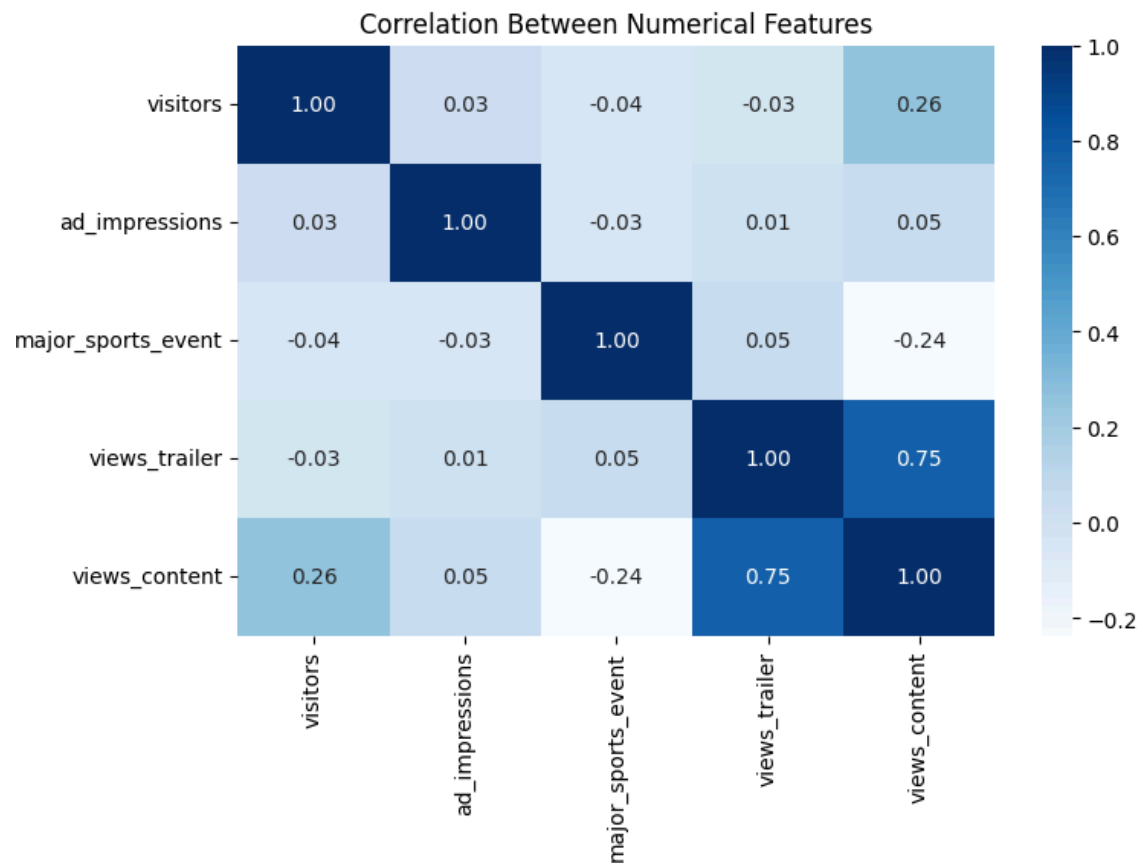


correlation between trailer views and content views: 0.753962214820536

3.5 Additional Insights

To go a step further, I also checked the correlation between numeric columns using a heatmap. But I had to drop the categorical columns for this because they can't be used in correlation directly.

- `views_content` was highly correlated with:
 - `views_trailer` (0.75)
 - `visitors` (also high)
- `major_sports_event` had a slight negative correlation, which means content released during sports events might get fewer views.
- `ad_impressions` also had some positive relation but could be highly related to `visitors`, which we had to check later for multicollinearity.



Summary of EDA

From all this analysis, I learned:

- Trailer views and platform visitors are very important indicators of success.
- Releasing content on Saturdays or in Summer might help improve performance.
- If a big sports event is happening on the release day, viewership might go down.
- Most content performs similarly, but genre and season can give small boosts.

This step helped me know which features to keep an eye on when making the model.

4. Data Preprocessing

In this step, I cleaned and prepared the dataset so that it can be used for building a machine learning model. This is a very important step because the model won't work properly if the data is not in the correct format.

4.1 Checking for Duplicate Rows

I checked if there were any duplicate rows in the dataset. Duplicate rows can affect model accuracy and should be removed.

Result:

There were 0 duplicate rows in the dataset, so I did not need to remove anything.

4.2 Encoding Categorical Variables

Some of the columns in the dataset were in text format. Machine learning models can't work with text directly, so I converted them using one-hot encoding.

The columns I encoded were:

- genre
- dayofweek
- season

This created new columns like genre_Comedy, dayofweek_Monday, season_Summer, and so on. Each of these new columns has a value of True or False depending on whether that category is present for a particular row.

After encoding, the shape of the dataset changed from (1000, 8) to (1000, 21) because many new columns were added.

4.3 Splitting the Data into Features and Target

Next, I separated the data into two parts:

- X – all the input features that will be used for prediction (this included columns like visitors, ad_impressions, views_trailer, and the new dummy variables for genre, dayofweek, and season).
- y – the target column that I want to predict, which is views_content.

Final shapes:

- Shape of X: (1000, 20)
- Shape of y: (1000,)

Now, the data was ready to be used in a machine learning model.

5. Model Building – Linear Regression

After preparing the data, I used a Linear Regression model to figure out which features are affecting the first-day views of content. Linear regression is a good starting model because it helps in understanding the contribution of each feature clearly.

5.1 Splitting the Data

To build and test the model, I split the dataset into two parts:

- 80% as training data (800 rows), which is used to train the model
- 20% as testing data (200 rows), which is used to evaluate how well the model performs on unseen data

Training set shape: (800, 20)

Testing set shape: (200, 20)

This split makes sure the model doesn't overfit and gives us an idea of how it performs on new data.

5.2 Training the Model

I used the training dataset to train a Linear Regression model. The model learns the relationship between the input variables (like visitors, ad impressions, trailer views, etc.) and the target variable (first-day views).

5.3 Coefficients and Intercept

After training, I checked the intercept and the coefficients of the model.

- Intercept: 0.0548
This means that if all features were 0, the model would predict around 0.0548 million views.
- Most impactful features (positive):

- visitors = 0.1289
- dayofweek_Saturday = 0.0526
- dayofweek_Wednesday = 0.0495
- dayofweek_Monday = 0.0451
- season_Summer = 0.0446

These features have the biggest positive impact on views.

- Least helpful or negative features:

- major_sports_event = -0.0596
- genre_Romance = -0.0014

This means that if there's a major sports event on the release day, it reduces views.

	Feature	Coefficient
0	visitors	0.128909
12	dayofweek_Saturday	0.052561
16	dayofweek_Wednesday	0.049532
11	dayofweek_Monday	0.045065
18	season_Summer	0.044605
13	dayofweek_Sunday	0.038818
15	dayofweek_Tuesday	0.032412
19	season_Winter	0.026532
17	season_Spring	0.023201
14	dayofweek_Thursday	0.019637
10	genre_Thriller	0.011518
5	genre_Drama	0.010636
9	genre_Sci-Fi	0.010008
6	genre_Horror	0.009434
7	genre_Others	0.004984
4	genre_Comedy	0.004389
3	views_trailer	0.002311
1	ad_impressions	0.000008
8	genre_Romance	-0.001385
2	major_sports_event	-0.059559

5.4 Training Accuracy (R-squared)

I checked the R-squared value of the model on the training data.

R-squared on training set: 0.7868

This means that about 78.68% of the variation in the first-day views is explained by the model. It shows that the model fits the data pretty well.

6. Testing the Assumptions of Linear Regression

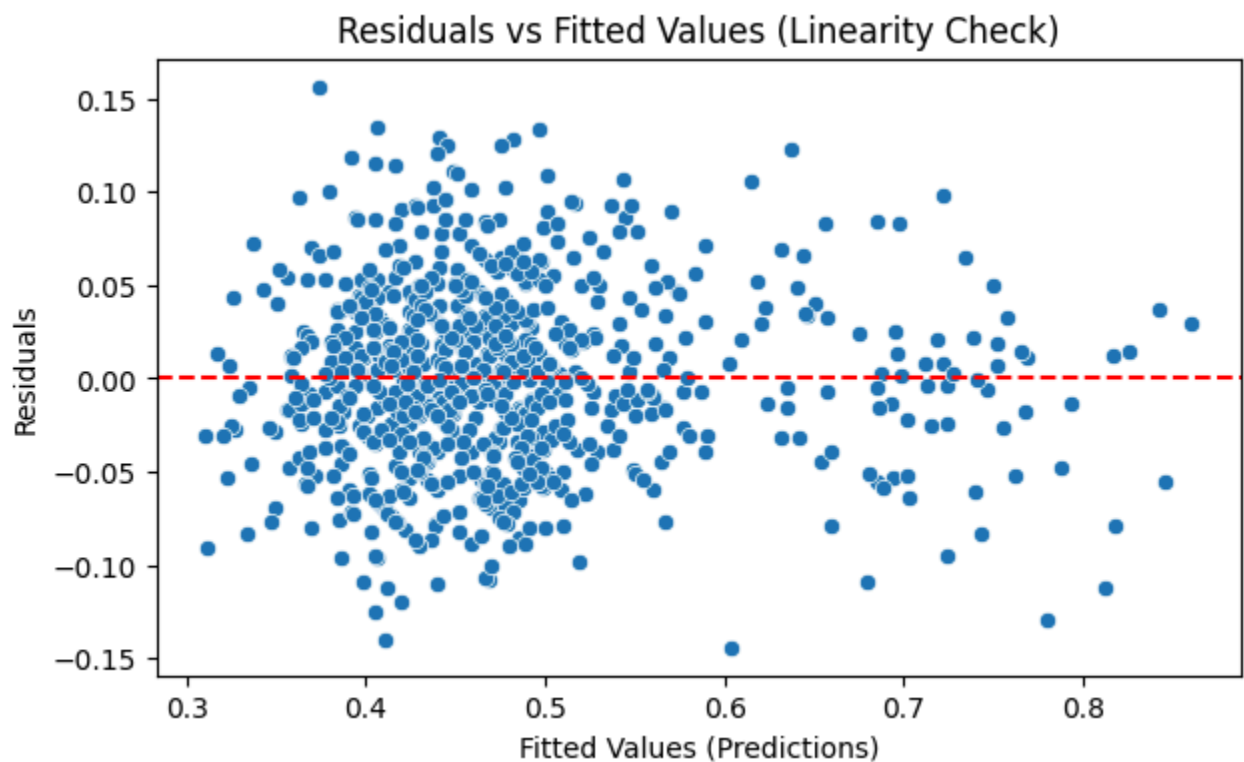
After building the model, I checked whether it satisfies the key assumptions of linear regression. These assumptions help ensure the model is accurate and reliable.

6.1 Linearity

To check if there is a linear relationship between the input features and the target variable, I plotted the actual values vs. the predicted values.

The plot showed that most of the points were close to a straight diagonal line, which means that the model captures a linear relationship well.

This assumption is satisfied.



6.2 No Multicollinearity

Multicollinearity means that some of the input features are highly related to each other. This can confuse the model.

To check this, I calculated the VIF (Variance Inflation Factor) for each feature.

Most features had VIF values below 5, which is fine. But two features had high VIF:

- ad_impressions: 15.18
- visitors: 10.52

This means these two variables are highly correlated and might be giving the model the same type of information.

This assumption is partially satisfied.

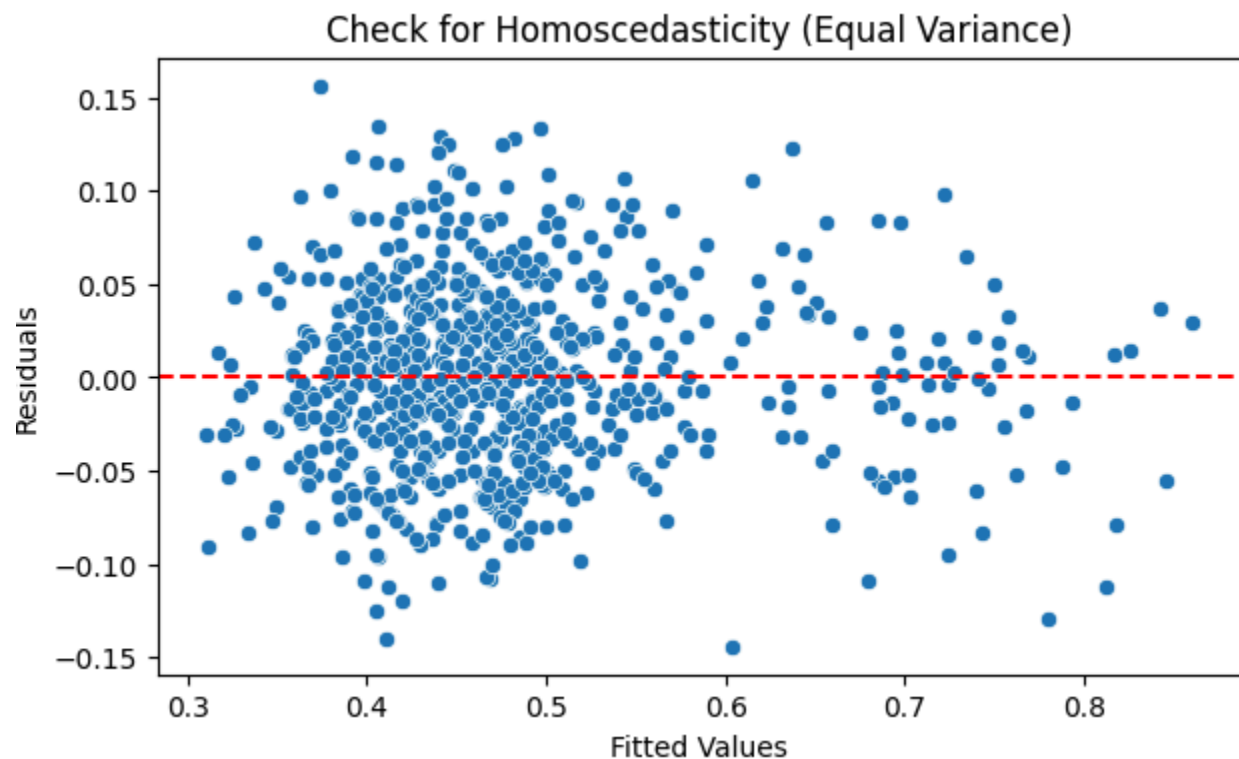
	Feature	VIF
1	ad_impressions	15.180256
0	visitors	10.521414
3	views_trailer	4.358402
7	genre_Others	2.960980
19	season_Winter	2.044921
18	season_Summer	1.966443
17	season_Spring	1.936544
4	genre_Comedy	1.905067
10	genre_Thriller	1.894294
5	genre_Drama	1.890737
9	genre_Sci-Fi	1.877466
16	dayofweek_Wednesday	1.875649
8	genre_Romance	1.814690
6	genre_Horror	1.789937
2	major_sports_event	1.707364
14	dayofweek_Thursday	1.255447
12	dayofweek_Saturday	1.242962
13	dayofweek_Sunday	1.190984
15	dayofweek_Tuesday	1.074365
11	dayofweek_Monday	1.071910

6.3 Homoscedasticity

Homoscedasticity means the errors (residuals) of the model should have constant variance across all levels of the predicted values.

I plotted the residuals (errors) against the predicted values. The points were randomly spread and didn't form a pattern, which shows that the variance is constant.

This assumption is satisfied.

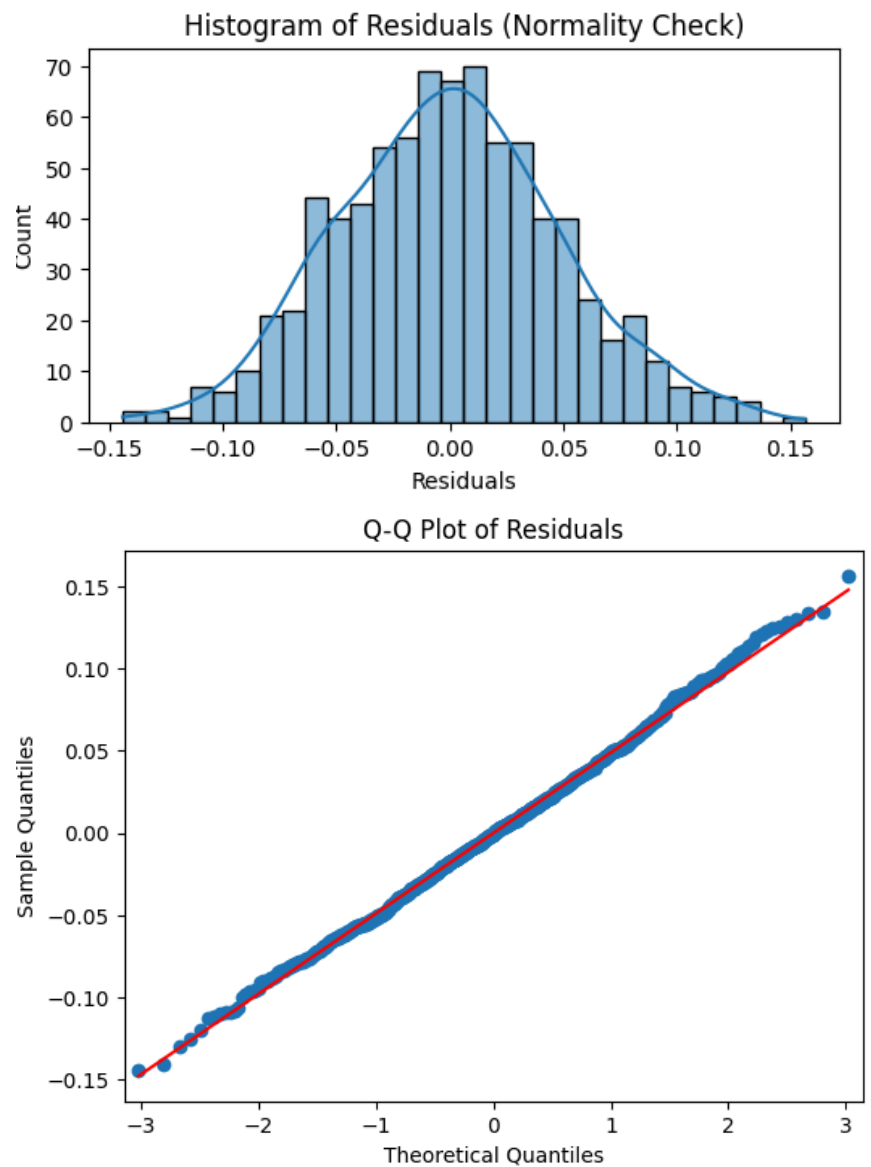


6.4 Normality of Errors

This assumption checks if the model errors follow a normal distribution. I checked this using two plots: a histogram and a Q-Q plot of residuals.

The histogram looked close to a bell-shaped curve, and the Q-Q plot showed the points mostly followed the diagonal line. This means the errors are approximately normally distributed.

This assumption is satisfied.



7. Model Performance Evaluation

I evaluated how well the model performs on the **test data**, which was not used during training. This helps to understand how the model will behave on unseen, real-world data. I used the following evaluation metrics:

7.1 Mean Absolute Error (MAE)

MAE tells how much the predictions are off from the actual values, on average.

- **MAE = 0.0399**
- This means the model's predictions are off by about 0.04 million views on average, which is quite small.

7.2 Mean Squared Error (MSE)

MSE gives the average of the squared errors. It punishes larger errors more than smaller ones.

- **MSE = 0.0025**

A lower MSE means better performance.

7.3 Root Mean Squared Error (RMSE)

RMSE is the square root of MSE. It is in the same unit as the target variable and easier to interpret.

- **RMSE = 0.0500**

So the average error is around 0.05 million views.

7.4 R-squared on Test Data

This tells how much of the variation in the target variable is explained by the model on unseen data.

- **R-squared (Test Set) = 0.7743**

So, the model is able to explain **77.43%** of the variation in the first-day views on test data, which is quite good.

Final Thoughts on Model Performance

The model performs well both on training and test data. The errors are small and consistent. Also, R-squared values for both training (78.68%) and testing (77.43%) are close to each other, which shows that the model is not overfitting.

8. Insights and Business Recommendations

After building the model and analyzing the results, I got some useful insights that can help the OTT platform (ShowTime) improve their first-day content viewership.

8.1 Important Features That Affect Views

1. Visitors

The number of visitors to the platform in the past week is the most important factor. More visitors usually means more views on release day.

Recommendation: Try to increase weekly visitors using promotions, discounts, or special content drops.

2. Day of the Week

Content released on Saturday, Wednesday, and Monday got better views.

Recommendation: Plan new and popular content releases on these days to get higher engagement.

3. Season

Releases in the Summer season got higher views compared to Fall or Spring.

Recommendation: Launch big shows or movies during summer for best results.

4. Major Sports Event

If a big sports event is happening on the same day, the views go down.

Recommendation: Avoid releasing content on days when there are IPL matches, World Cups, etc.

5. Genre

Most genres like Thriller, Drama, Sci-Fi, Horror had a small positive effect. Romance had a slightly negative effect.

Recommendation: Focus on promoting genres that work better, and maybe rethink how Romance content is presented.

6. Ad Impressions and Trailer Views

These features had only a small effect.

Recommendation: It may help to focus on improving the quality of ads and trailers instead of just increasing numbers.

8.2 Final Suggestions

- Learn more about user interests to release the right content at the right time.
- Try different release days and genres using small experiments (A/B testing).
- Collect more data over time to build even better models in the future.

Conclusion

In this project, I worked on a dataset from an OTT platform called ShowTime to understand what factors affect the number of views a content gets on its first day of release. I started with Exploratory Data Analysis (EDA) to study the dataset and answer some key questions about viewership patterns.

After that, I prepared the data by handling encoding and checking for duplicates. Then, I built a Linear Regression model to predict the first-day content views using different input features like visitors, day of the week, season, and genre.

The model performed well, with an R-squared value of 0.7868 on training data and 0.7743 on test data, which means it was able to explain most of the variation in viewership. I also tested the model on important assumptions like linearity, normality, and multicollinearity to make sure it was valid.

In the end, I got some helpful insights like the importance of choosing the right release day, avoiding major sports events, and focusing on increasing platform visitors.

This kind of analysis can help OTT platforms make better business decisions, plan content releases smartly, and improve user engagement.

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